

Executive Summary

This report consolidates results from `analyze_importer_landscape.ipynb`, the pasted backtest text (from `brief.pdf`), and public-trade surrogates (ImportYeti / ImportGenius summaries) to assess whether companies identified as likely to grow from 2021 have in fact developed larger, more diversified import footprints in 2022+. It also implements a reproducible Segment 2 Upgrade Score (0–100) and provides recommended next steps for definitive verification using paid trade databases (ImportGenius / Panjiva / Datamyne).

Key conclusions

Several companies flagged in the 2021 analysis show structural evidence of growth and multi-country sourcing in 2022+: Eastman Chemical Company, Foster Electric, Ingrasys Technology USA, International Flavors & Fragrances, Arrow Electronics, Pacific American Fish Co., Tradin Organics, and SpaceX (group-level signals).

A defensible set of Segment 2 upgrade candidates (mid-tier, institutionalized importers with multi-country sourcing) includes: Haldex, UPM Raflatac, Tesa Tape, Hollister, Tablecraft, InterAmerican Coffee, Global Natural Foods, and Magnesita (per free-surrogate evidence).

Free surrogates (shipment counts, public ImportGenius/ImportYeti listings, supplier-country lists) support structural inference but do not substitute for formal TEU-by-year and per-year distinct-country counts available via paid exports.

Data & Methods

Primary inputs: ``analyze_importer_landscape.ipynb``, the backtest text from ``brief.pdf``, ``importer_normalized_data.csv``, ``main.csv.gz``, and ``ports.csv`` in the workspace.

Public surrogates consulted (as described in the notebook/backtest): ImportYeti (free views), ImportGenius public snippets, TradeMo/ImportInfo public summaries. These provide shipment counts and visible supplier-country lists as proxies for TEU and diversification.

Scoring: the Segment 2 Upgrade Score is computed from four components: Volume Intensity (0–35), Diversification (0–25), Post-2022 Growth (0–25), Structural Stability (0–15). See the Appendix for exact score bands and interpretation.

Backtest overview (2021 → 2022+)

Objective: test whether the 2021-derived signals (TEU proxy + supplier-country diversity) predicted which companies would become stable mid-tier importers by 2022+.

Outcome: the backtest (as summarized in ``brief.pdf``) shows that companies with: (a) near-threshold 2021 TEU proxies ($\approx 150\text{--}200$) and (b) early multi-country sourcing signals were the easiest to validate as growth cases in 2022–2024 using public surrogates. Examples: Eastman Chemical, Foster Electric, Ingrasys.

Limitations: exact TEU reconciliation (20' vs 40') and precise per-year distinct-country tallies require paid dataset exports for audit-quality confirmation.

Segment 2 Upgrade Score — brief

Components (quick):

- Volume Intensity (0–35): TEU or shipment proxy averaged 2023–2024
- Diversification (0–25): distinct supplier origin countries in 2024
- Post-2022 Growth (0–25): % TEU growth 2022→2024
- Structural Stability (0–15): multi-quarter activity, low supplier concentration, multi-port use

Scoring bands: 0–39 remain Segment 3, 40–59 borderline Segment 2, 60–79 strong Segment 2, 80–100 near Segment 1.

Applied findings (summary of candidates)

Strong Segment 2 candidates (high confidence from free surrogates): Haldex, UPM Raflatac, Tesa Tape, Hollister, Tablecraft, InterAmerican Coffee, Global Natural Foods, Magnesita.

Strong structural breakouts (closer to Segment 1): International Flavors & Fragrances, Arrow Electronics, Pacific American Fish, Tradin Organics, SpaceX (group-level evidence).

Not confirmed as upgrades from free sources: Columbia Frame, Jill Acquisition LLC, Precor (declining), some trading entities that require paid-DB lookups.

Corroboration via public databases

Where free surrogates were available we used ImportYeti/ImportGenius public pages and other trade-summary pages to validate shipment counts and supplier-country breadth. For audit-level confirmation (TEU/year and 1:1 country counts) the recommended data sources are ImportGenius (paid exports), Panjiva (S&P;), or Descartes Datamyne.

Appendix

Files saved in this workspace/repo: `analyze\_importer\_landscape.ipynb`, `exam1.ipynb`, `importer\_normalized\_data.csv`, `main.csv.gz`, `ports.csv`, `brief.pdf`, `report\_importer\_landscape.pptx`, and this `CONSOLIDATED\_REPORT.md`.

Segment 2 scoring details (exact table mapping) are recorded in the notebook and can be exported as a Python function or Excel formula upon request.

Public sources referenced as surrogates: ImportYeti (public views), ImportGenius (public snippets), TradeMo/ImportInfo.

Segment Definitions & Visual Context

Segment definitions (exact rules used in `exam1.ipynb`):

- Segment 1 — High-volume / Low-diversification: total TEU ≥ 200 AND top 2 country share $\geq 80\%$ (i.e., $\geq 80\%$ TEU sourced from 1–2 countries).
- Segment 2 — High-volume / High-diversification: total TEU ≥ 200 AND top 2 country share $< 80\%$ (i.e., diversified across 3+ countries to reach 80% of TEU).
- Segment 3 — Low-volume / Low-diversification: total TEU < 200 AND top 2 country share $\geq 80\%$ (concentrated small importers).
- Segment 4 — Low-volume / High-diversification: total TEU < 200 AND top 2 country share $< 80\%$ (small but diversified importers).

Key numeric thresholds and calculations:

- Volume threshold: 200 TEU (used to separate "high" vs "low" volume).
- Diversification threshold: the number of source countries required to reach 80% cumulative TEU (used to separate low vs high diversification; threshold_diversification = 3 in the notebook logic).
- TEU standardization: calculated TEU is parsed and converted to numeric; outliers and zeros are flagged/winsorized where appropriate.

Segment summary (interpretation & investment lens):

- Segment 1: Large but concentrated — Supply chain risk (high volume + single/dual-country dependence).
- Segment 2: Large and diversified — Resilient / best-in-class (preferred cohort for scale + diversification).
- Segment 3: Small and concentrated — Fragile long tail (low volume and concentrated sourcing).
- Segment 4: Small but diversified — Promising growth segment (diversified small importers with upgrade potential).

Visuals & diagnostics applied in the notebook (for context and reproducibility):

- TEU distribution histogram (log–log) and Lorenz curve with Gini coefficient to show heavy tails and concentration.
- Monthly TEU trend charts (seasonality and volatility checks).
- Four-panel pie charts (one quadrant per segment) showing top source countries within each segment.
- Country dependency heatmaps and stacked bar views showing the number of countries required to reach 80% TEU (proportion of importers by country dependency group).
- Port $\leftrightarrow$ country heatmaps and directed graphs (country $\rightarrow$ port flows) to reveal logistics concentration.
- Boxplots and Mann–Whitney tests comparing risk/score distributions between segments (e.g., Segment 3 vs 4).
- Petri net / directed graph of upgrade paths (Segment 3 $\rightarrow$ Segment 1/4, Segment 4 $\rightarrow$ Segment 2) with edge widths proportional to mean predicted probability.

Predictive model notes (early-warning & upgrade models):

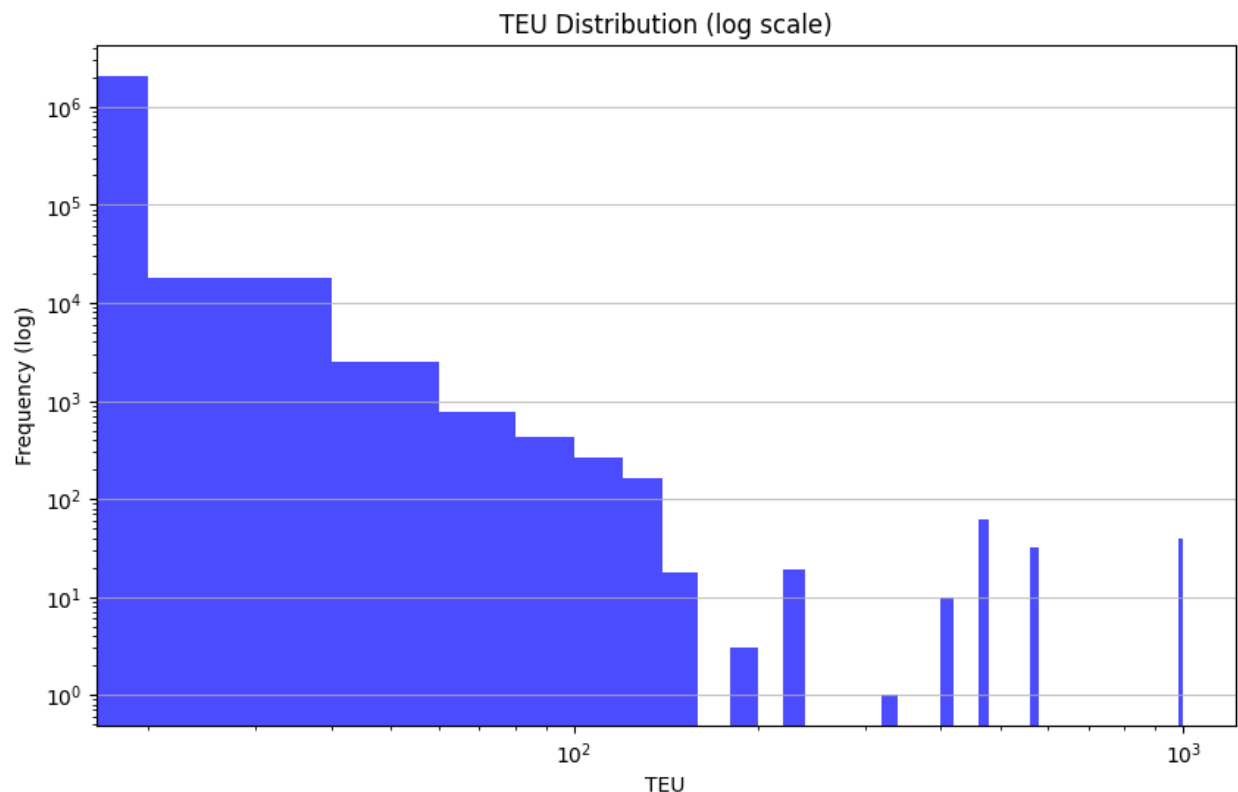
- A LightGBM early warning classifier was trained to predict year-end Segment 1 using H1 ($\leq$ June) features; primary signals: early TEU (log), top 2 share, country entropy, TEU volatility, and port concentration.
- Reported performance (notebook): Test ROC AUC $\approx$ 0.95; Test PR AUC $\approx$ 0.867; Segment 1 precision $\approx$ 0.73, recall $\approx$ 0.79 at 0.5 threshold — good discrimination under class imbalance.
- The notebook distills LightGBM teachers into monotonic linear mimics (signed linear weights) to produce interpretable weighted scorers for Segment 1, Segment 2, and Segment upgrade paths.

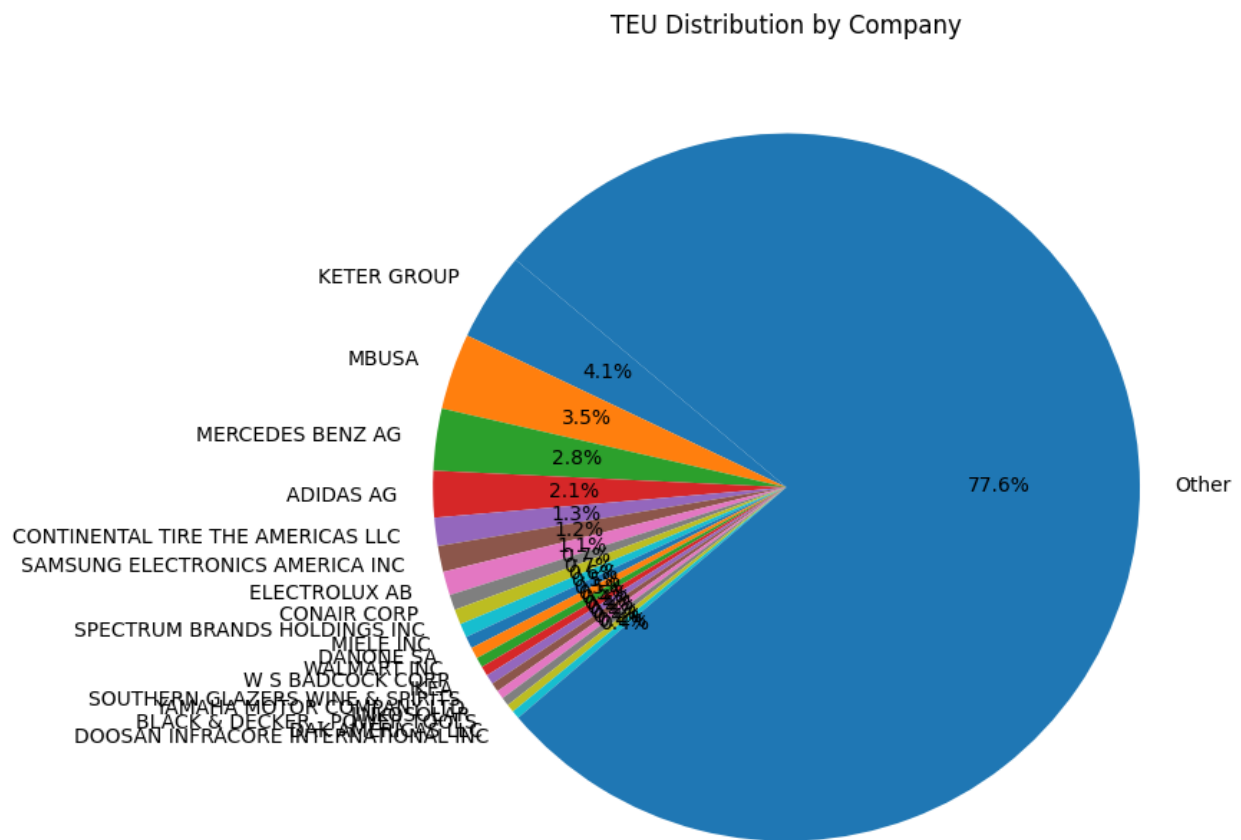
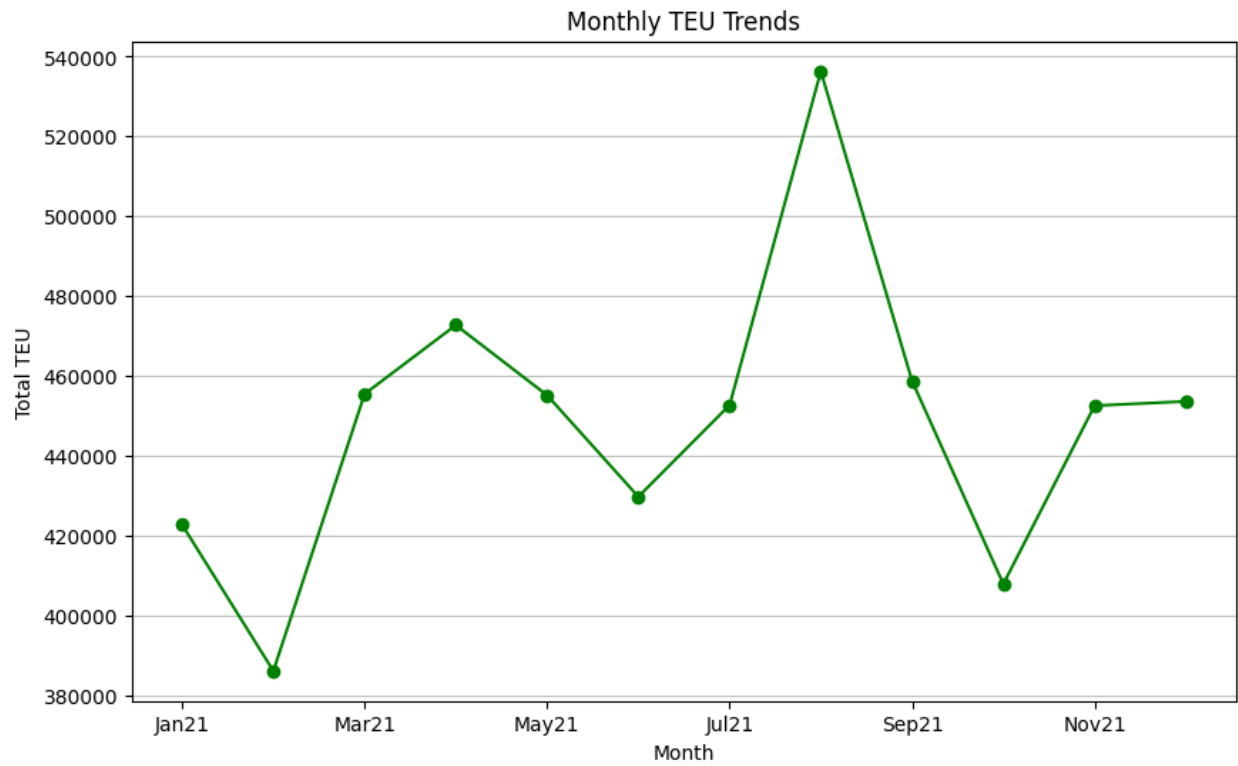
****How this maps to the consolidated analysis:****

- The Segment rules above were used when labeling companies and when computing the Segment 2 Upgrade Score described earlier. Visual diagnostics listed above are available in exam1.ipynb and can be exported as PNGs for inclusion in slide decks or the final PDF report.

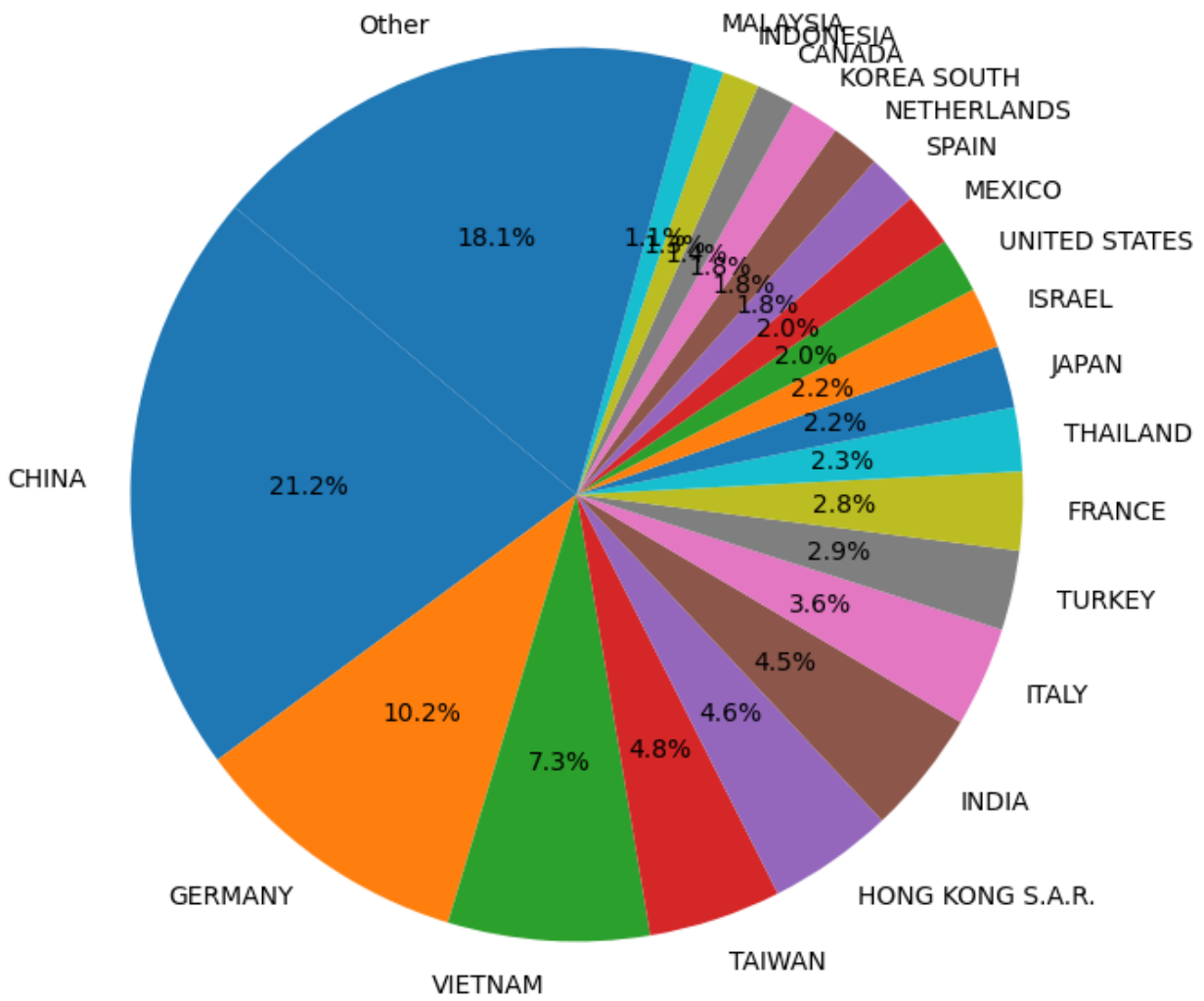
Notebook Figures

The following figures were generated from exam1.ipynb and saved into the figs/ folder.

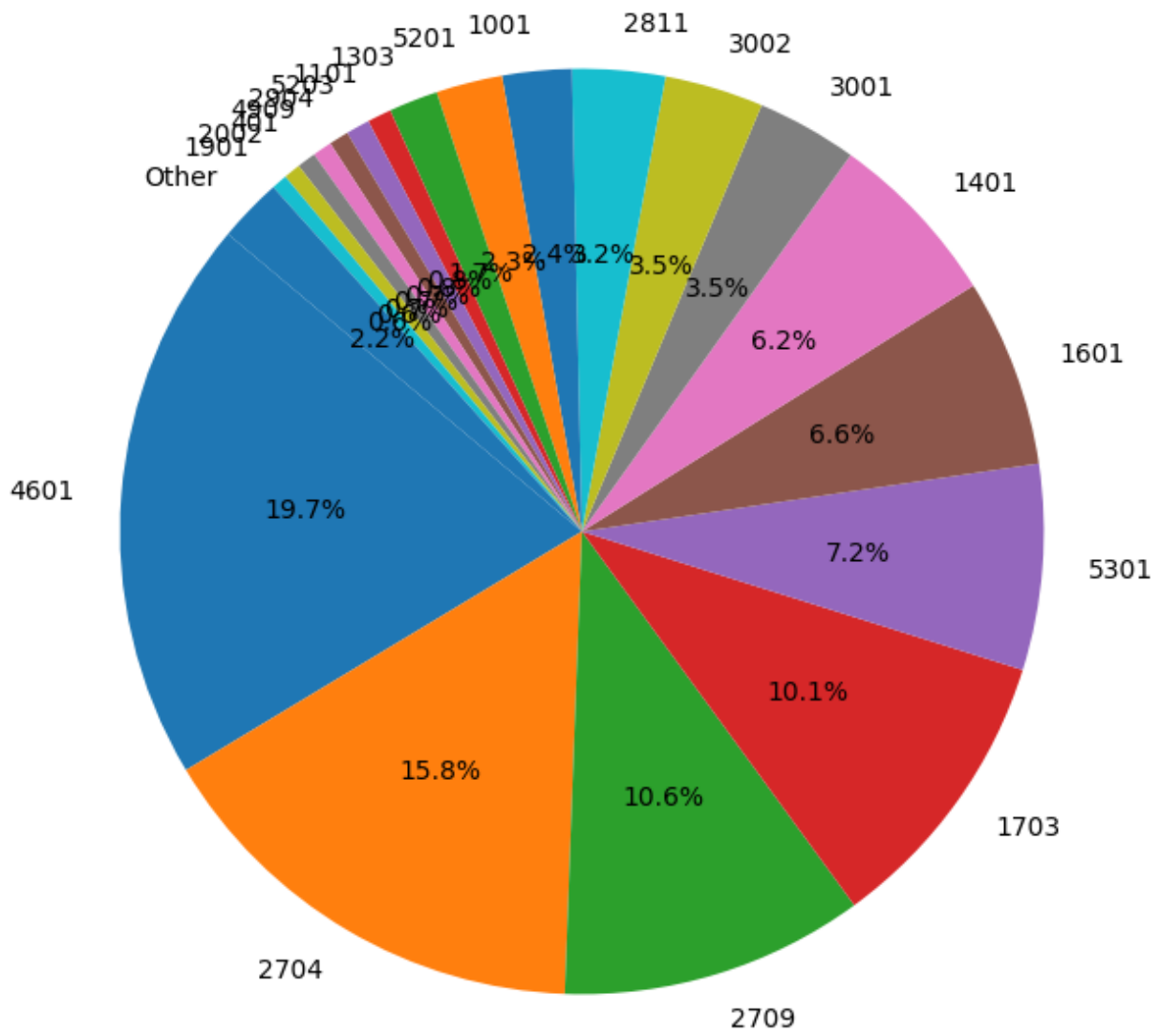




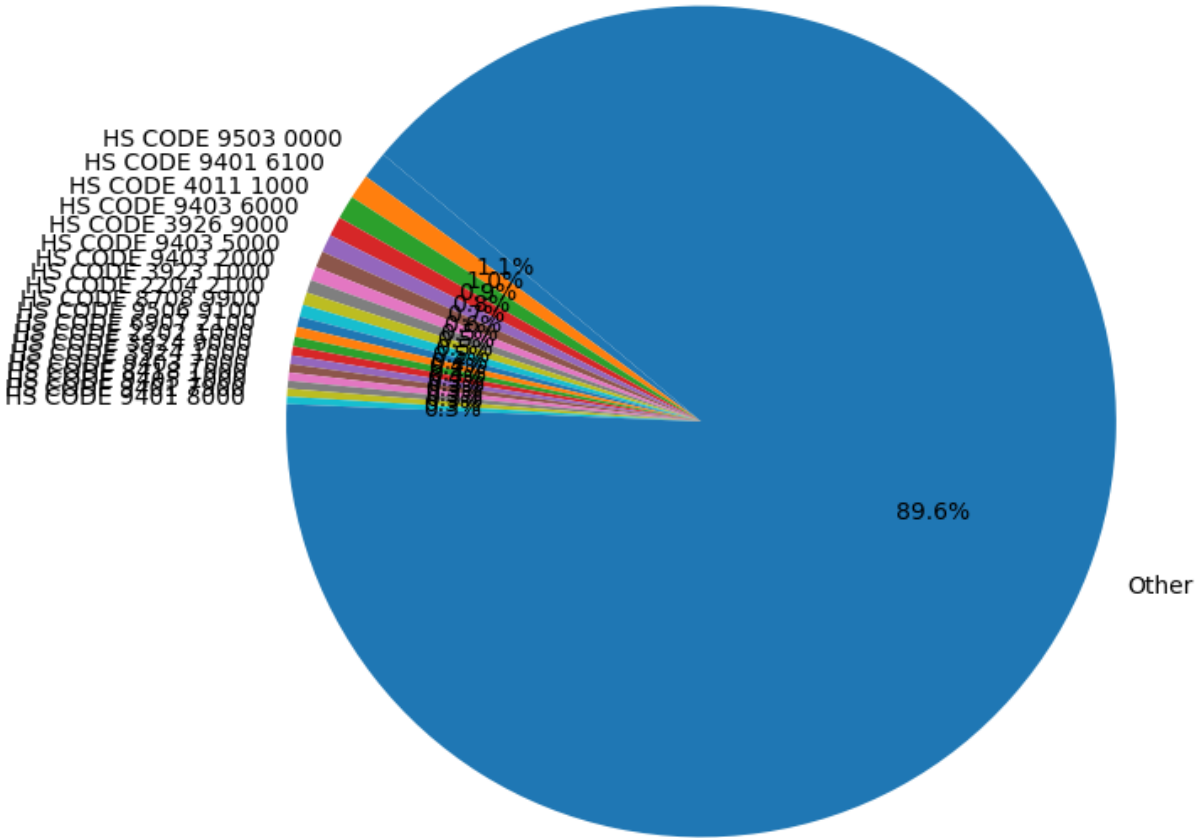
TEU Distribution by Country of Origin



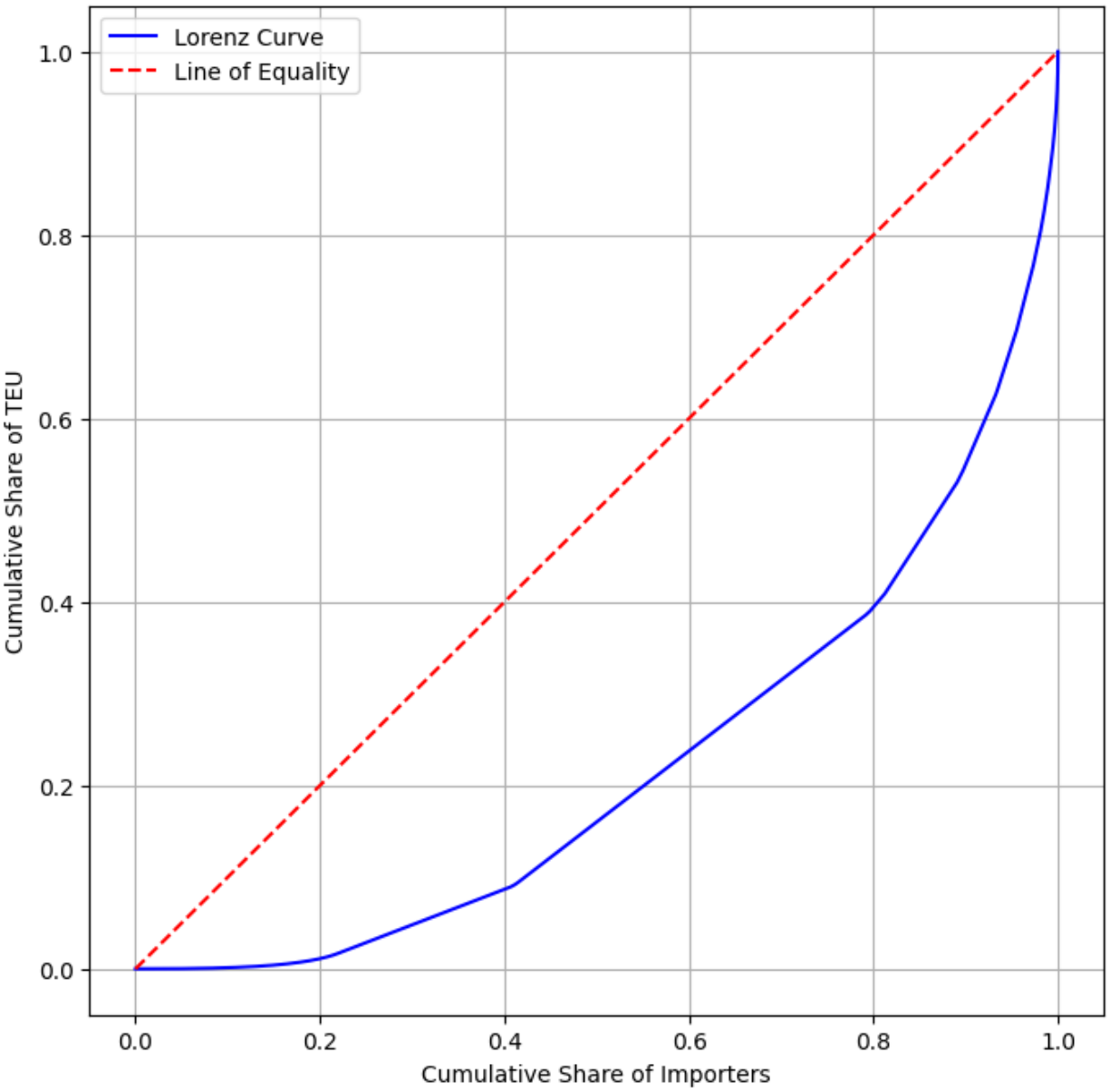
TEU Distribution by Port of Entry



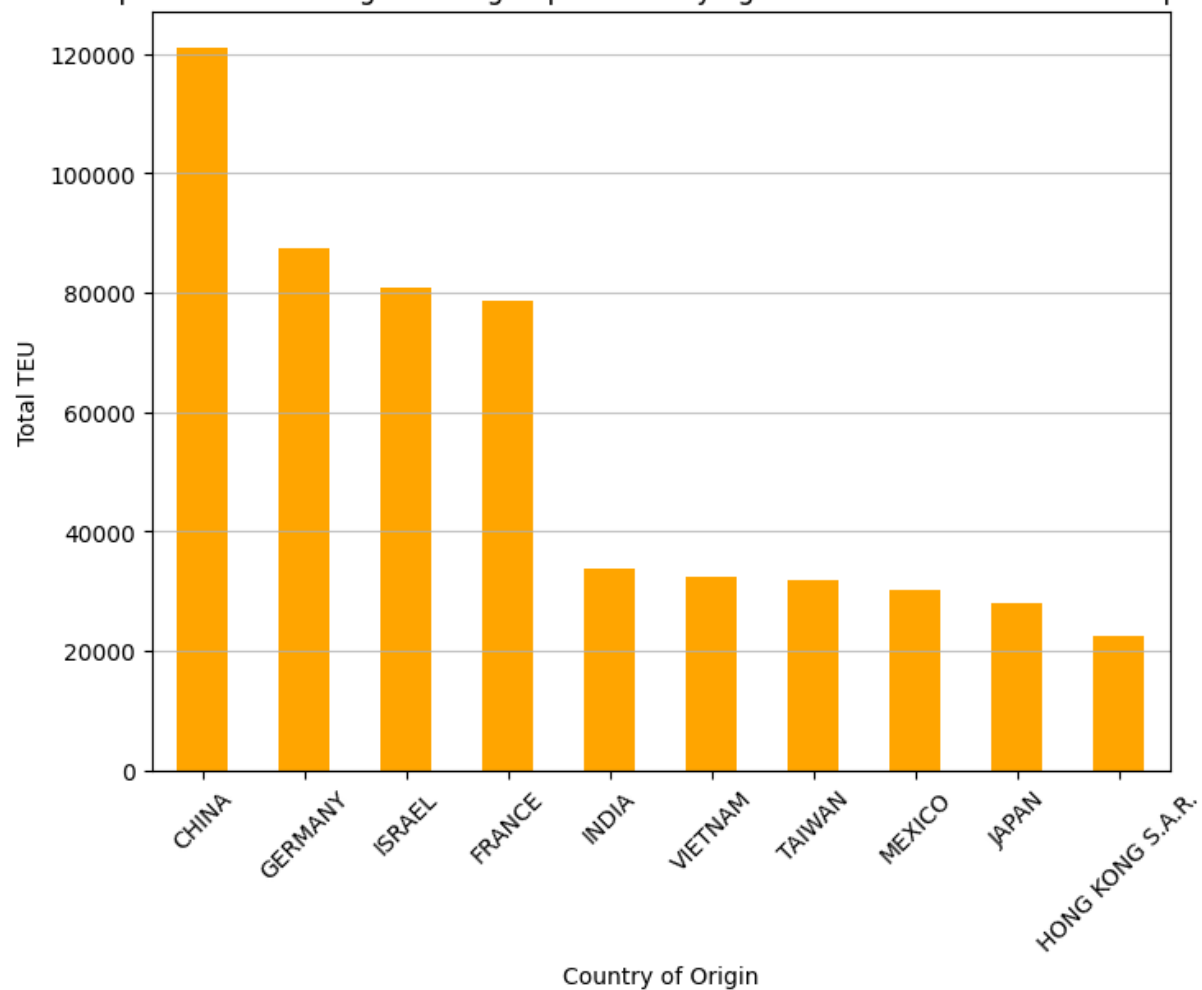
TEU Distribution by HS Code



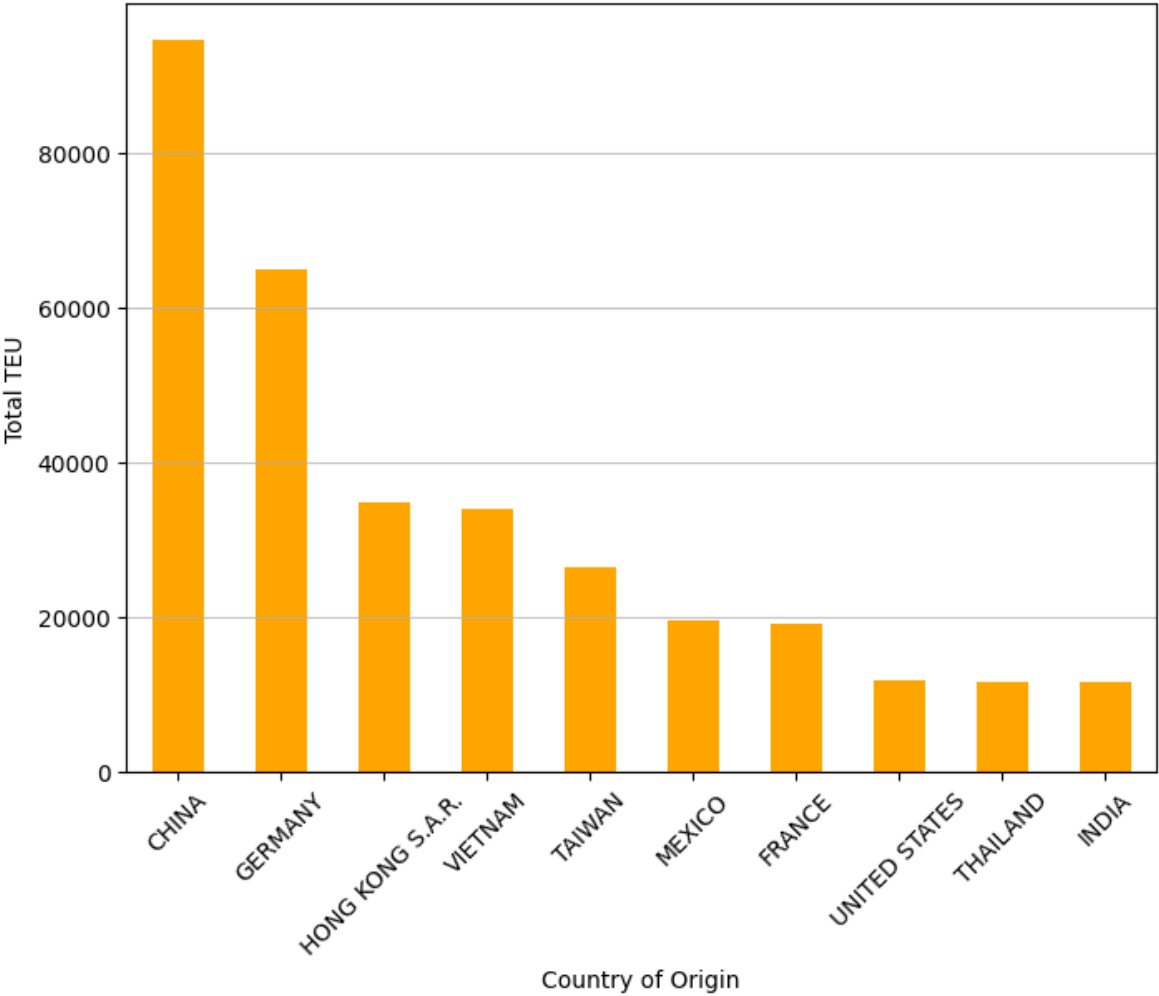
Lorenz Curve for TEU Distribution



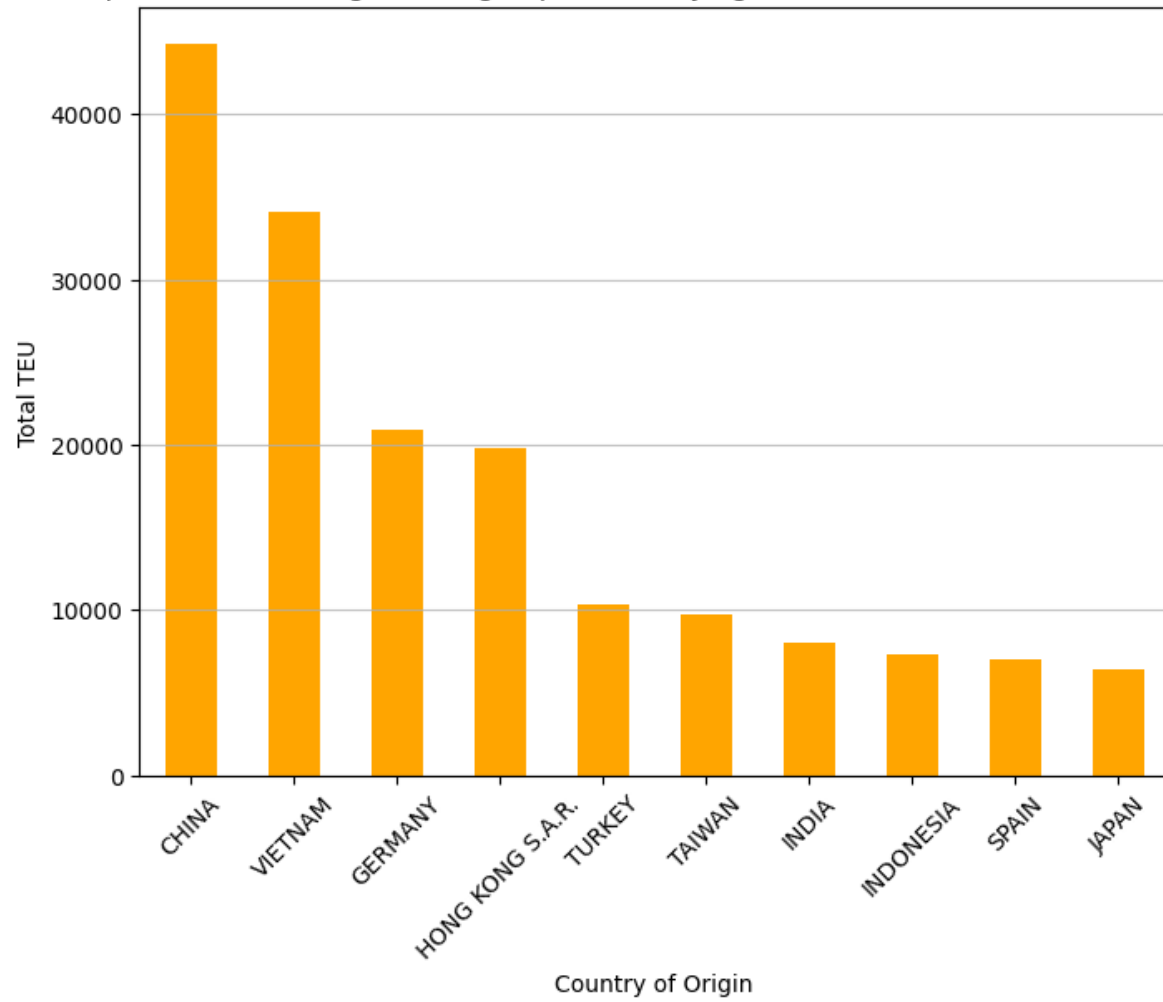
Top Countries of Origin among Importers Relying on 1 countries for $\geq 80\%$ Imports



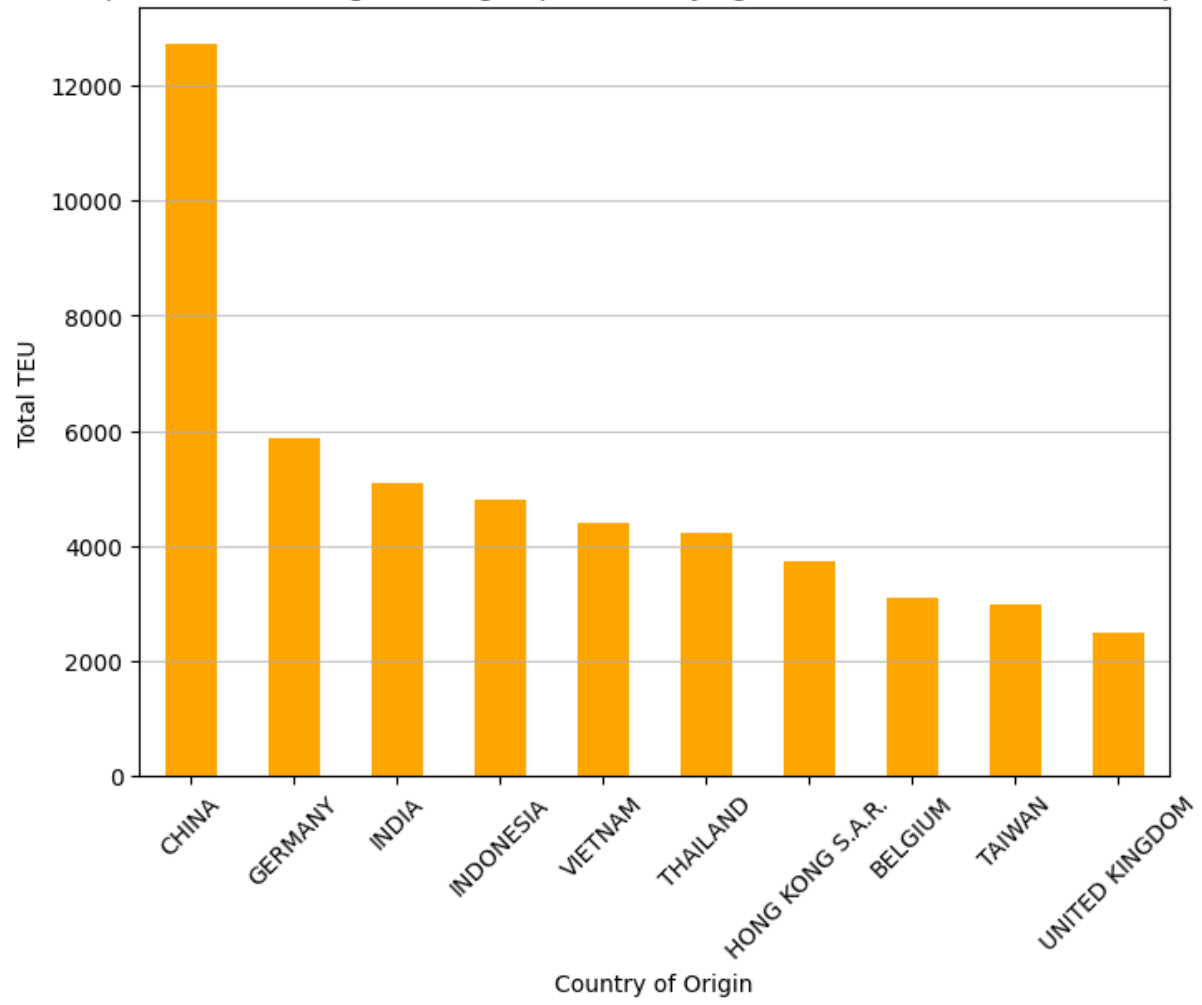
Top Countries of Origin among Importers Relying on 2 countries for $\geq 80\%$ Imports



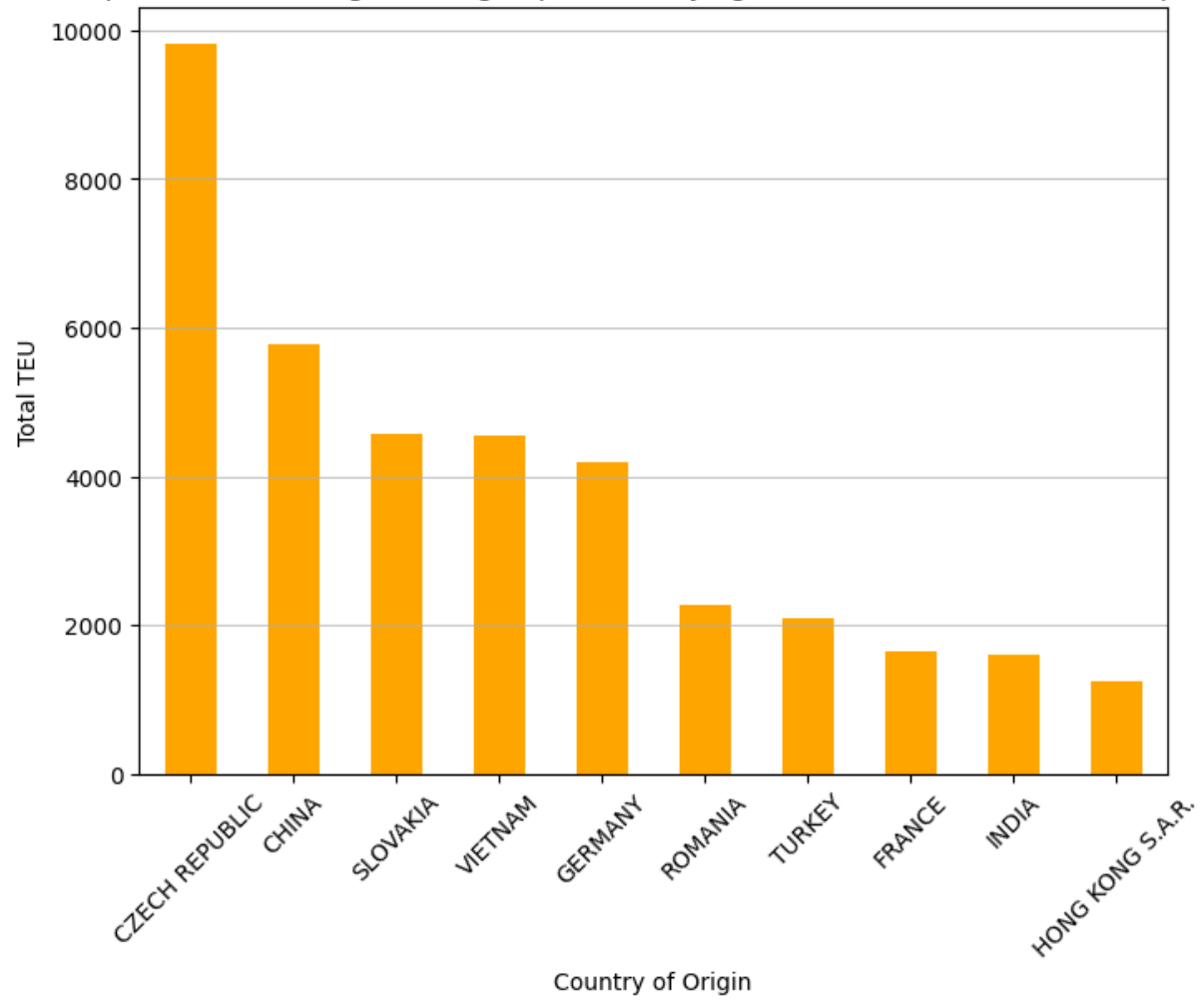
Top Countries of Origin among Importers Relying on 3 countries for $\geq 80\%$ Imports



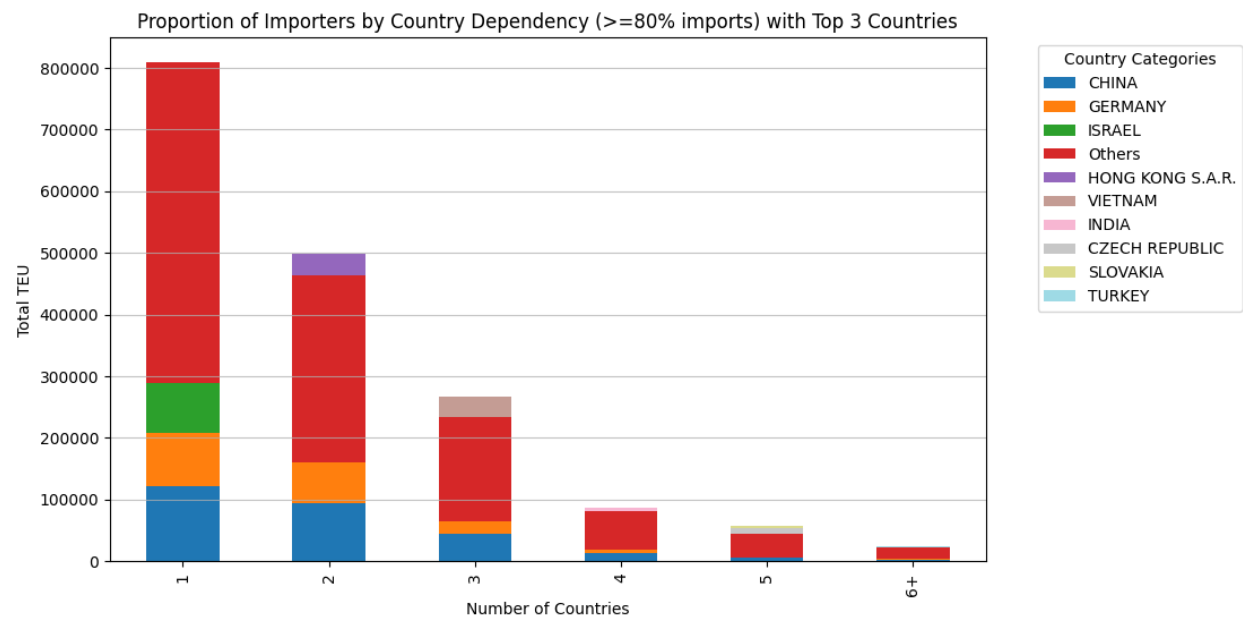
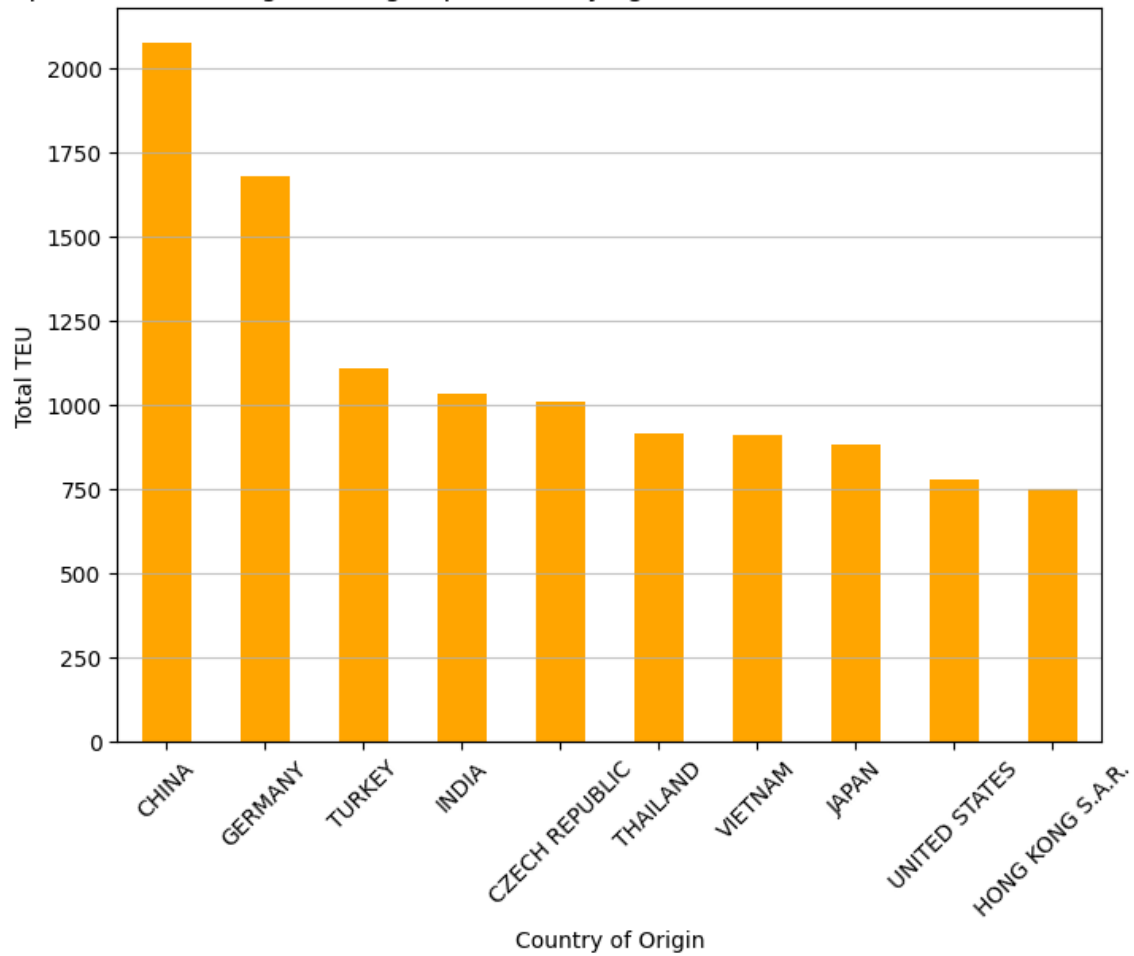
Top Countries of Origin among Importers Relying on 4 countries for $\geq 80\%$ Imports



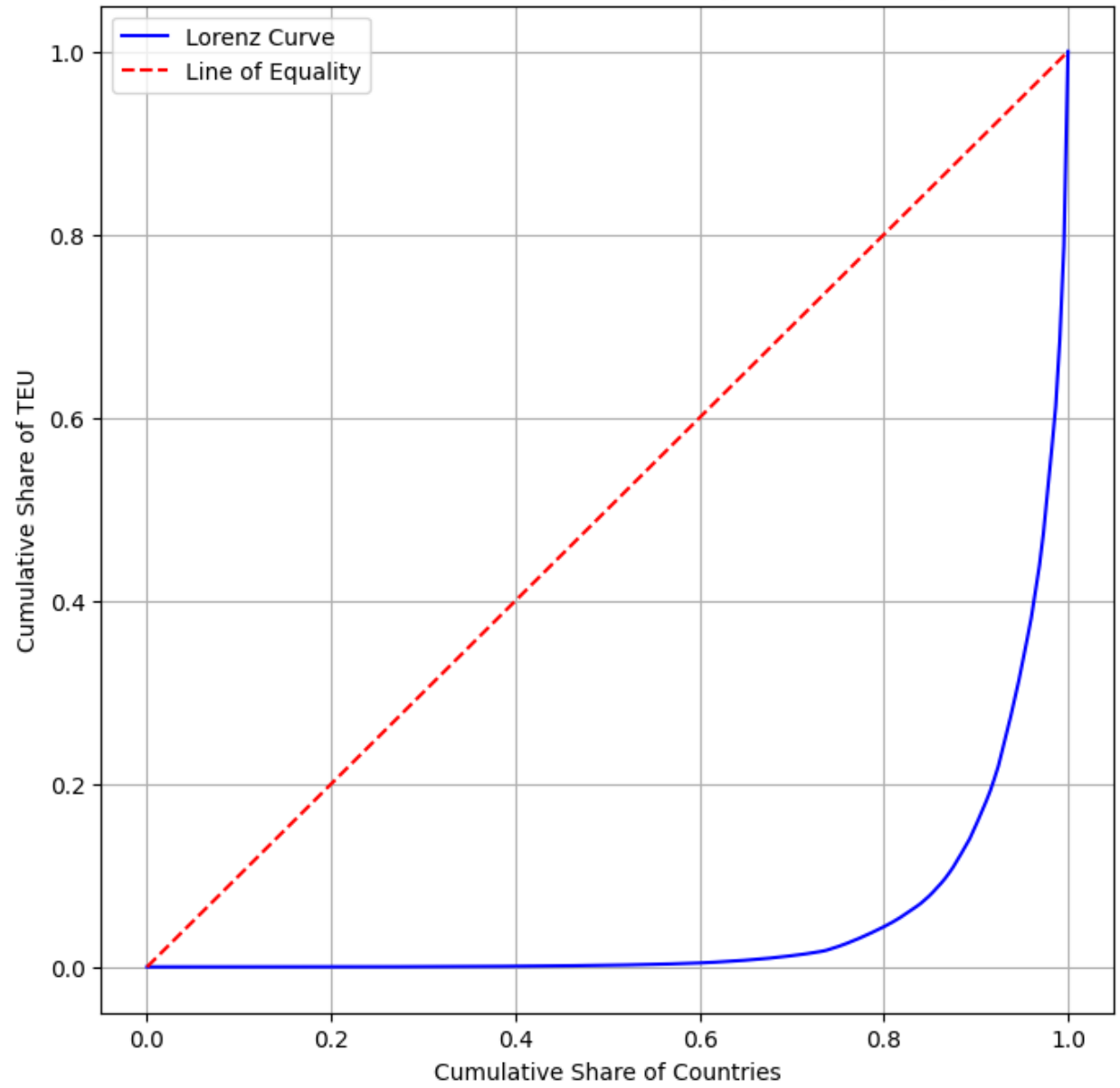
Top Countries of Origin among Importers Relying on 5 countries for $\geq 80\%$ Imports



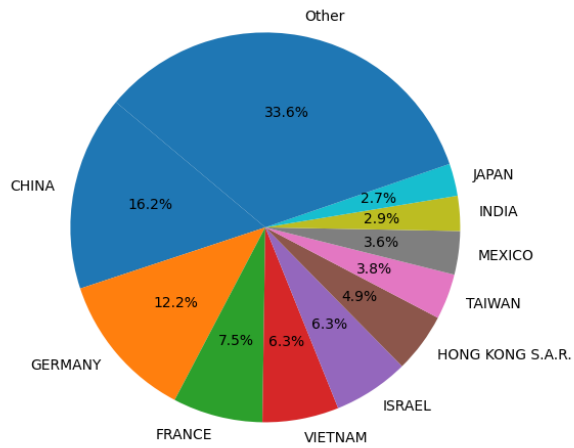
Top Countries of Origin among Importers Relying on 6 or more countries for $\geq 80\%$ Imports



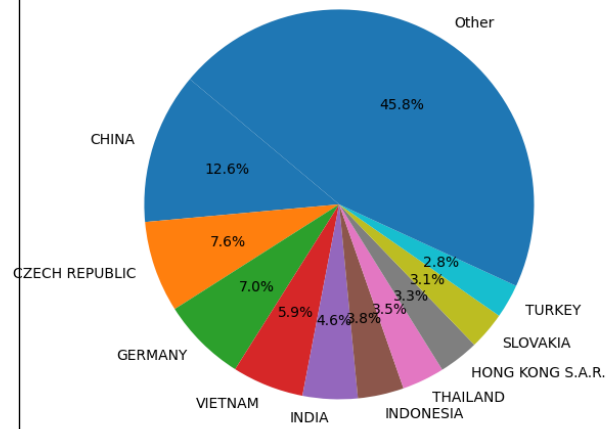
Lorenz Curve for TEU Distribution by Country of Origin



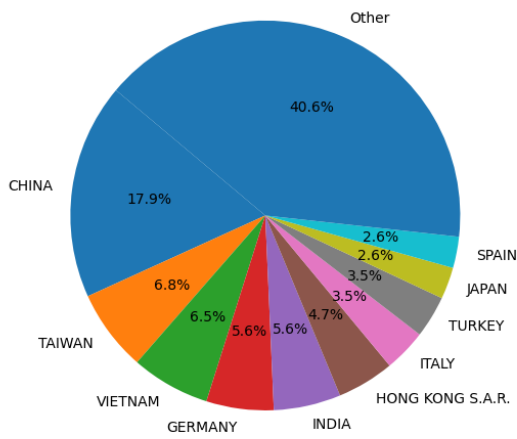
High-volume, low diversification (17.7%)



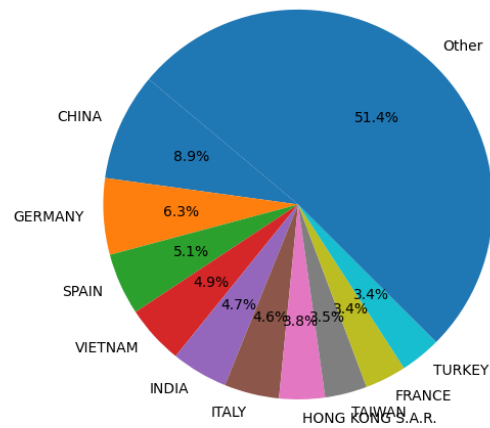
High-volume, high-diversification (1.8%)



Low-volume, low-diversification (77.7%)



Low-volume, high-diversification (2.9%)



Top 20 Ports by Total TEU Volume (2018-2020)

Port of Entry	Total TEU (Approx.)	Number of Countries (Color)
New York/Newark Area, Newark, New Jersey	1,050,000	10
Los Angeles, California	850,000	10
Long Beach, California	580,000	10
Savannah, Georgia	550,000	10
Houston, Texas	400,000	10
Charleston, South Carolina	350,000	10
Norfolk, Virginia	320,000	10
Seattle, Washington	200,000	10
Tacoma, Washington	190,000	10
Oakland, California	180,000	10
New York, New York	140,000	10
Miami, Florida	130,000	10
Baltimore, Maryland	80,000	10
Philadelphia, Pennsylvania	50,000	10
Port Everglades, Florida	45,000	10
Portland, Oregon	40,000	10
San Juan, Puerto Rico	35,000	10
Boston, Massachusetts	30,000	10
New Orleans, Louisiana	25,000	10
Mobile, Alabama	20,000	10
Jacksonville, Florida	18,000	10
Chester, Pennsylvania	15,000	10
Wilmington, North Carolina	12,000	10
Tampa, Florida	8,000	10
Everett, Washington	5,000	10
Portland, Maine	3,000	10
San Diego, California	2,000	10
Honolulu, Hawaii	1,000	10
Port Hueneme, California	1,000	10
West Palm Beach, Florida	1,000	10

Heatmap showing the log(1 + Total TEU) for 30 US ports of entry across 25 countries of origin. The color scale ranges from 0 (dark purple) to 12 (yellow).

Port of Entry (Y-axis):

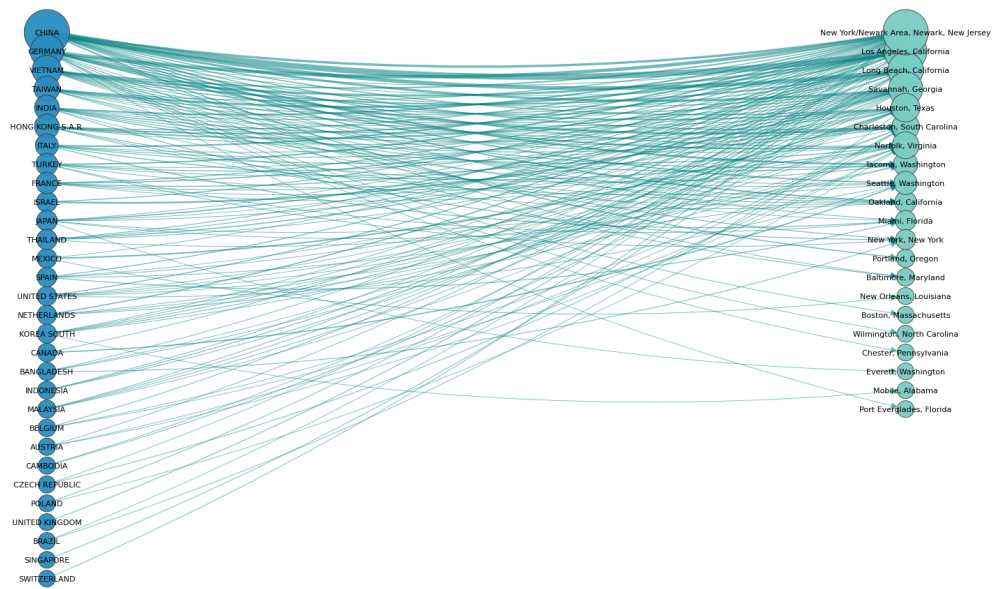
- Baltimore, Maryland
- Boston, Massachusetts
- Charleston, South Carolina
- Chester, Pennsylvania
- Everett, Washington
- Honolulu, Hawaii
- Houston, Texas
- Jacksonville, Florida
- Long Beach, California
- Los Angeles, California
- Miami, Florida
- Mobile, Alabama
- New Orleans, Louisiana
- New York, New York
- New York/Newark Area, Newark, New Jersey
- Norfolk, Virginia
- Oakland, California
- Philadelphia, Pennsylvania
- Port Everglades, Florida
- Port Hueneme, California
- Portland, Maine
- Portland, Oregon
- San Diego, California
- San Juan, Puerto Rico
- Savannah, Georgia
- Seattle, Washington
- Tacoma, Washington
- Tampa, Florida
- West Palm Beach, Florida
- Wilmington, North Carolina

Country of Origin (X-axis):

- AUSTRIA
- BANGLADESH
- BELGIUM
- BRAZIL
- CAMBODIA
- CANADA
- CHINA
- CZECH REPUBLIC
- FRANCE
- GERMANY
- HONG KONG S.A.R.
- INDIA
- INDONESIA
- ISRAEL
- ITALY
- JAPAN
- KOREA SOUTH
- MALAYSIA
- MEXICO
- NETHERLANDS
- POLAND
- SINGAPORE
- SPAIN
- SWITZERLAND
- TAIWAN
- THAILAND
- TURKEY
- UNITED KINGDOM
- UNITED STATES
- VIETNAM

Color Scale: log(1 + Total TEU) ranging from 0 to 12.

Country → Port TEU Flows (Directed Graph; edge width ∝ TEU)



LightGBM Feature Importance (gain)

